

Newcastle United

Technical Appendix & Additional Findings

Supporting analysis for the Newcastle United performance analytics portfolio

Match-level analysis: 43 matches | Event-level analysis: 38 Premier League matches, 962 shots, 108 goal events

1. Purpose and scope

This appendix preserves the statistical detail and secondary findings behind the main portfolio. It is intended to complement the one-page executive summary and full report rather than replace them. The early match-level work uses a 43-match sample spanning the available 2025/26 and early 2026/27 data, while the later event-level work focuses on the complete 38-match 2025/26 Premier League season.

The analysis is explanatory and descriptive. In-match variables such as shots-on-target differential and half-time state should not be interpreted as pre-match predictors, and small player/pair samples should be treated as signals for further video review rather than causal effects.

2. Shot quality and shots-on-target control

The strongest simple relationship with points in the 43-match sample was shots-on-target (SOT) differential. Raw shot differential was much weaker, supporting the hypothesis that Newcastle's ability to generate and suppress shots on target is more informative than total shot volume alone.

Variable	Spearman correlation with points
SOT differential	+0.653
Newcastle shots on target	+0.533
Shot conversion	+0.431
Shot accuracy	+0.383
Raw shot differential	+0.164

SOT thresholds

Condition	Matches	PPG	Win rate
SOT differential ≥ 0	24	2.00	58.3%
SOT differential < 0	19	0.47	10.5%
SOT differential $\geq +2$	12	2.58	83.3%

These thresholds provide a more intuitive football interpretation than a correlation alone: Newcastle's results were substantially stronger when they at least matched the opponent for shots on target.

3. Venue and game-state patterns

Venue	Matches	PPG	Win rate	Shot diff	SOT diff
Home	22	1.68	50.0%	+0.55	+0.55
Away	21	0.95	23.8%	-1.76	-0.67

The home/away difference was visible not only in results but also in the underlying shooting profile, with Newcastle moving from positive average shot and SOT differentials at home to negative differentials away.

Half-time state	Matches	PPG	Win rate
Leading	18	1.11	33.3%
Drawing	12	1.75	41.7%
Trailing	13	1.23	38.5%

This early half-time table motivated deeper investigation rather than a causal conclusion. The relationship between half-time state and final result had a Cramer's V of approximately 0.36 in this sample.

4. Adjusted win model

A regularised logistic regression was used to examine whether SOT differential remained informative after accounting for venue, opponent pre-match PPG, half-time state and raw shot differential. Features were standardised and model performance was assessed with repeated stratified five-fold cross-validation.

Model	Mean AUC
Venue + opponent strength	0.703
+ half-time game state	0.795
+ raw shot difference	0.767
+ SOT difference	0.861
Full model	0.873

Factor	Adjusted direction	Relative strength
SOT differential	Strong positive	Largest effect
Opponent pre-match PPG	Negative	Strong
Raw shot differential	Negative once SOT difference is controlled	Moderate
Trailing at half-time	Negative	Moderate
Home advantage	Positive	Moderate
Leading at half-time	Positive	Smaller

In the fitted full model, a one-standard-deviation increase in SOT differential corresponded to roughly 4.1 times the odds of winning. This is a standardised model coefficient, not a claim that one additional shot on target quadruples win probability.

The full model is explanatory rather than a pre-match forecasting model because SOT differential and half-time state are observed during the match. The 43-match sample is also small, so effect sizes should be interpreted as evidence worth investigating rather than definitive laws.

5. What drives SOT differential?

Shot-volume differential and shot-accuracy differential together explained 86.9% of the observed variation in Newcastle's SOT differential in-sample. This decomposition suggests that SOT control can be reached through different combinations of volume and accuracy.

Context	SOT diff	Shot diff	Accuracy diff	PPG
Home	+0.55	+0.55	-0.84pp	1.68
Away	-0.67	-1.76	-1.74pp	0.95
Lower-strength opponents	-0.06	+1.56	-7.78pp	1.88
Higher-strength opponents	-0.80	-4.00	+4.27pp	0.93
Leading at HT	+0.50	-2.06	+8.15pp	2.00

Drawing at HT	+1.17	+1.58	-0.72pp	1.75
Trailing at HT	-1.60	-0.73	-11.77pp	0.27

Against stronger opponents, the larger constraint appears to be shot volume, whereas against lower-strength opponents Newcastle generated more shots but did not consistently convert that volume into SOT dominance. When trailing at half-time, both SOT control and shot accuracy deteriorated.

6. Event-level post-goal findings

The event-level phase covers all 38 Premier League matches, 962 shots and 108 goal events, using added-time-adjusted 5-, 10- and 15-minute windows.

15-minute window	xG differential	SOT differential
After Newcastle scored	+0.160	+0.190
After Newcastle conceded	-0.062	-0.125

After scoring, Newcastle's shots averaged 0.170 xG compared with 0.090 for opponents. Newcastle's average shot distance was approximately 13.8m versus 15.8m for opponents.

Post-score venue	PGCI	Average pressure signal
Home	55.6	+9.16
Away	45.8	+2.79

The Post-Goal Control Index (PGCI) summarises Newcastle's share of shots, shots on target and xG in the window, with 50 representing even control.

7. Territorial pressure as a control mechanism

Post-score pressure	PGCI	SOT differential	xG differential
Positive territorial pressure	62.4	+0.519	+0.314
Negative territorial pressure	32.9	-0.400	-0.118

The clearest event-level association is that Newcastle retained much stronger SOT and xG control when the post-goal territorial-pressure signal remained positive. The effect is particularly relevant away from home, where the average pressure signal and PGCI were both lower. This supports a football interpretation centred on retaining territorial control rather than simply counting shots.

Territorial pressure is a proxy for match pressure/territory, not possession percentage. These associations do not prove that Newcastle deliberately defend deeper after scoring.

8. Player and unit signals for video review

Player	Complete windows	PGCI	xG diff	SOT diff
Fabian Schär	13	66.8	+0.332	+0.85
Lewis Miley	17	64.7	+0.277	+0.88
Jacob Murphy	16	61.7	+0.258	+0.69
Tino Livramento	16	60.5	+0.211	+0.63
Bruno Guimarães	33	53.4	+0.184	+0.33

Pair	Shared windows	PGCI	xG diff	SOT diff
Miley + Tonali	8	80.3	+0.524	+1.50
Miley + J. Murphy	8	77.2	+0.353	+1.13
Schär + J. Murphy	6	70.6	+0.331	+1.00
Schär + Gordon	9	68.4	+0.304	+0.78
Schär + Livramento	8	68.3	+0.331	+0.75

These player and pair results are descriptive. Opposition strength, venue, score state, team selection and season timing are substantial confounders. They are best used to select sequences for tactical video review, with questions around midfield spacing, counter-press positioning, full-back height, second balls and progression after regains.

9. Interpretation and limitations

- Association is not causation: the analysis identifies patterns linked with outcomes and control, not proof of tactical intent or player causality.
- The 43-match match-level sample is small, and the event-level player/pair windows are smaller still.
- SOT and half-time variables occur during matches, so the adjusted win model is explanatory rather than a pre-match prediction model.
- PGCI and territorial pressure are constructed analytical measures and should be interpreted alongside match video and tactical context.
- The strongest use of the player-combination results is hypothesis generation for video review rather than selection recommendations.

10. Portfolio takeaway

Across the match-level and event-level work, the consistent theme is that Newcastle's control is better described by shot quality, shots-on-target control and territorial pressure than by raw shot volume alone. The most actionable next step is targeted video analysis of the situations and player combinations associated with retaining positive territory after scoring, particularly in away matches.